

Immersion, Flow and Usability in video games

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ABSTRACT

Context: Nowadays, video games look and feel increasingly realistic, and provide engaging and immersive experiences to players. Understanding what makes games so appealing and immersive is important to grasp in order to provide better experiences.

Objectives: To provide a better understanding of the concepts of flow, immersion and usability in video games and understand how usability interact with these different concepts.

Method: 20 participants took part in user tests and answered a survey about usability and immersion.

Results: Usability had a positive effect on immersion which was fully mediated by appeal. Learnability was correlated with absorption by activity, and usability was correlated with fluency of performance.

Conclusions: Usability and appeal are both contributing to the feeling of immersion which in turn increases the general appreciation of the game. Immersion and flow are both relevant to understand players' appreciation of the game.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **HCI design and evaluation methods**; • **Laboratory experiments**;

KEYWORDS

Usability, Immersion, Flow, User Testing, Games

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1 INTRODUCTION

1.1 Engagement and immersion

Atmospheric, immersive, realistic, deep, addictive. . . All these terms are frequently used to designate the level of immersion in a game, which contributes to players' enjoyment and engagement.

O'Brien and Toms [44] defined engagement as a concept associated with aesthetics, novelty, involvement, usability, and the evaluation of the salience of the experience. Game engagement and immersion are important factors to consider when designing a

video game, especially nowadays with countless accesses to many leisure activities, and innumerable games to play. Engagement and immersion are sometimes used interchangeably by different authors, which can be an issue as the same word ends up being used for distinct concepts and different words are used for describing the same concept [1]. Addiction and engagement have also been used synonymously in the past, but they are inherently different and should not be assimilated to the same concept [12]. Engagement is a multidimensional construct composed of cognitive, behavioral, and affective components [56], which arises from the interaction between the player and the game. Engagement in video games is often associated with different factors such as enjoyment, motivation, flow, fun, arousal, and immersion itself [1, 11]. For Brown & Cairns [11], immersion is a subcomponent of that influences engagement itself. Different papers have studied the concept of immersion in video games and brought out components that define it. Jennett et al. [31] for instance, extracted five factors from their factor analysis and named them: cognitive involvement, real-world dissociation, control, challenge, and emotional involvement. Brockmyer and their colleagues [9], as well as Banyte & Gadeikiene [4] described game-engagement in four dimensions: immersion, flow, presence, and absorption, whereas Procci et al. [47] used an almost identical model but replaced absorption with involvement. All these studies have used similar concepts to learn about the player experience in games and to measure game engagement. They also used game engagement and immersion interchangeably. Ermi & Mäyrä [20] extracted three main components from their qualitative study: sensory immersion, challenge-based immersion, and imaginative immersion. This model was also supported by a factor analysis with a larger sample of adult players ($n = 234$). Brown and Cairn [11] for their part, were the first authors to study the concept of immersion in video games and defined it as the degree of involvement with a game. Their findings suggest there are three distinct levels and each one has its barriers to attain an immersive experience. These three levels are engagement, engrossment, and total immersion.

Consequently, engagement refers to the first level of immersion in the sense that players have to invest time, energy, and attention in the game first to immerse themselves. From this perspective, engagement allows the players to be drawn into the game. It is also important to note that the level of engagement is closely related to the players' preferences and tastes. That explains why various research has been done to find connections between game preferences and personality [49, 58]. To reach a certain level of engagement, players have to be invested first. Once the players are engaged in the game, they can decide to continue playing and the interest will thus increase. However, there is no emotional attachment to the game yet at this first level of immersion. Surprise and novelty are factors that can also help keep engagement and curiosity high by renewing motivations to play [30]. Moreover, Takatalo et al. [52] connected curiosity to involvement, and they describe it as

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the desire to find something new in a game. Another study [29] also showed that children's engagement with interactive exhibitions was influenced by three key elements: participation, which was defined as a playful process during which the children could have an impact on the information presented; narration (stories and linear structure), and co-presence of others. This corroborates Brown and Cairns' findings [11], as it shows that involvement and participation in a task can have an impact on the engagement level.

Engagement, presence, and cognitive absorption are concepts commonly used to describe the engaging experience (closely related to immersion). However, Jennett et al. [31] proposed that immersion is different from these concepts. Immersion and cognitive absorption share similar dimensions such as a feeling of control, attention, enjoyment, temporal dissociation and curiosity [2], but immersion is the experience of playing a video game whereas, cognitive absorption is more of an attitude towards the game. Immersion overlaps with flow as well, but flow represents an extreme case of immersion as it is supposed to be regarded as an optimal experience. Immersion allows more nuanced experiences during which the player can be immersed but still aware of its surroundings at the same time. Another difference is that a game can provide a highly immersive experience without meeting all criteria for attaining flow [41].

It is important to note that, having to perform a task accentuates the feeling of immersion thus, it is central in video games which often allow a high level of interactivity [46]. In one of the few factorial experiments done in video games, Christou [14] proposed from their results that immersion is a consequence of appeal. Players create their first impression based on the aesthetic dimensions of the game itself and then they decide to become immersed or not. This effect was both presents in experienced and inexperienced players which indicates that the level of experience did not influence the immersion level.

1.2 Flow and presence

Flow theory in games refers to the game capacity to immerse the player to the point of losing self-consciousness and track of time or any other external pressure by giving clear goals, direct feedback, and implicating challenges [13, 19, 41]. Flow is supposedly a consequence of a challenge-skill balance and the absorption in the task [22, 53], but other authors have found that dynamic pacing in which the skill-demand oscillates resulted in a greater level of flow when playing video games. The dynamic condition in which demand fluctuated but didn't surpass skill caused the highest enjoyment level [6]. Therefore, a perfect challenge-skill balance might not always be the preferred solution in games.

Presence is defined by Patrick et al. [46] as "the extent to which a person's cognitive and perceptual systems are tricked into believing they are somewhere other than their physical location" (p. 479). It is a feeling of existence in another environment. Jennett et al. [31] defined presence as a state of mind in which the player is put whereas immersion is an active experience. Weibel & Wissmath [54] also differentiated presence from flow. Presence is the sense of being in the game world, whereas flow is the feeling of being the actor of the actions in the game (a feeling of agency). Therefore, their point of view differs from the one of Brown and Cairns [11] who consider presence synonymous with the highest level of immersion.

Vividness and interactivity in games are thought to generate presence [8]. Other factors can impact presence such as atmosphere (graphic and sound quality) as well as attention (the more attentional sources and intense they are, the greater the presence will be) and empathy with the characters and the situation [8, 11, 51]. Witmer and Singer [57] also mentioned that feedback, concentration, and a sense of control define the degree of presence which are also considered part of the flow concept [18], and immersion. Nevertheless, presence is not synonymous with immersion. As for the link between immersion and flow, Michailidis et al. [40] have not found a clear difference between flow and immersion both from a behavioral and neurophysiological standpoint. But, the same authors also determined that presence appears to be different from immersion and, therefore should be different from flow as well.

Bachen et al. [3] concluded from their study, using a serious computer game, that presence was indeed a good predictor of flow, identification, and "game-specific empathy", which confirms what was stated earlier. Pre-experiment empathy level also influenced game-specific empathy and presence level which indicates that personality can shape the emotions felt during gameplay as well as the level of presence, and as a consequence influences flow and identification. Multiple authors have found results that show that presence in games is linked to games and more precisely a predictor of it [3, 32, 35, 43, 45, 52, 55]. Takatalo et al. [52] created the Presence-Involvement-Flow Framework (PIFF) to describe gaming experiences and they defined presence and involvement as "the switch between the real world and a game" (p.31).

1.3 Usability

Not much experimental research focusing strictly on the usability components has been done in video games. In Christou's research [15] participants were asked to play a game and then to answer a questionnaire. He found out that four out of the seven pragmatic attributes of the Attrakdiff questionnaire (simple, clear, innovative, manageable) correlated with the general appeal of the game. Their results showed that a link between usability and appeal is also present in video games as it is in other forms of interactive products. People who perceived the game usable also perceived it to be good and likable. These results go in the same direction as previous works attesting the effect of usability on perceived appeal [25, 27].

Immersion is also closely related to appeal in games [15], and players can have different motivations to play games which can then impact immersion [23, 58, 59]. It also has been shown that low usability leads to frustration [16, 21, 33], which in turn leads to lower flow level, as explained by Clough [16] who identified a flow-frustration dichotomy in computer user interface usage.

Video games put the players in a goal mode, and consequently both pragmatic and hedonic qualities shape the experience [28].

1.4 Research contribution

The setup of the study, in a game studio testing facility, allowed players to feel more at ease than in a university laboratory setting. It also allowed the players to play the same game for several days without much interruption, whereas often user tests last for several hours only. We could make sure players had never played the game

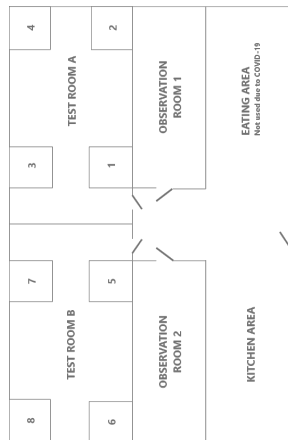


Figure 1: Plan of the lab. One-way mirrors separate the test rooms and observations rooms. The Numbers represent the eight stations.

since the game was unreleased at the time of the tests which removes the eventual effect of experience with a game. Most players also signed up for the test mainly in the interest of playing new games and not to obtain monetary compensation, which can be the case for studies that take place on campus with students. The sample was representative of the average video game player (age and hours played per week). Players were also selected based on their genres and playing habits so that they would most likely have similar player's motivations and profiles.

It also seems to be the first study to explore the link between usability, flow, and immersion in video games, to our knowledge.

2 METHOD

This study aims to investigate the effect of usability on immersion and flow. Immersion and flow are both impacted by the feeling of control and attention which can be enhanced with good usability [4, 9, 31, 47]. As previous research showed, bad usability can cause frustration which can negatively impact flow and immersion [16, 33, 39].

Participants took part in a playtest at a video game studio Lab (Figure 1).

They played a third-person action-adventure open world game that contains driving and shooting/combat mechanics. The game was released on PC, PS4 and Xbox One, but the tests were done on a pre-launch version of it, while still being under a strict NDA. They played it on PC with an Xbox controller. They played the same game for several days for a total of roughly 15 hours. In total, three test sessions took place with multiple participants at the same time, each playing the game at their own station. The participants didn't know each other and the recruitment criteria were: being familiar with playing games with a controller (PS4 or Xbox); considering open-world games and/or Role-Playing Games (RPG) one of their favorite genres; having played at least 10 hours in some other similar games (Figure 2); be 18 years old or older.

A post-test survey was administered to assess the perceived level of immersion, flow and usability. Based on previous research

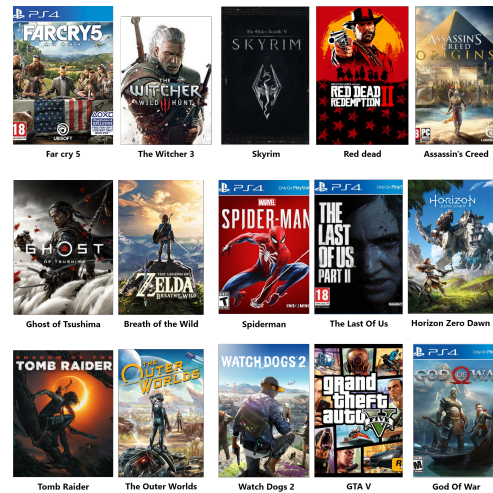


Figure 2: List of games the participants had to be familiar with, which are also similar to the game they played

on appeal, usability and [14, 25, 27], we can expect that appeal will mediate the relationship between usability and immersion.

2.1 Hypotheses

- Perceived usability is correlated positively with the perceived appeal
- Perceived usability increases the subjective experience of immersion
- Perceived usability increases the experienced flow state level
- Perceived usability effect on immersion is mediated by appeal

2.2 Questionnaire development

The questionnaire was composed of four sections and only minor adjustments were made to make it fit the topic when it was needed.

- The System Usability Scale (SUS) was used to assess the perceived usability of the game. It consists of 10 statements on a five points Likert scale [10].
- The 7 items from the attractiveness section of the Attrakdiff were used to measure hedonic qualities of the game [26]. Items are measured on a seven points semantic differential scale.
- The Immersive Experience Questionnaire (IEQ) [31] contains 31 statements that are assessed on a 5 points scale. The questionnaire is based on previous work from Agarwal & Karahanna [2] and Brown & Cairns [11].
- The Flow Short Scale (FSS) [48] was used to assess the flow state. The FSS is composed of 10 statements rated on a seven points Likert scale.

Lastly, the last question of the questionnaire was used to assess general appreciation of the game "On a scale from 1 to 10, how much did you enjoy the game?".

2.3 Participants

20 console players took part in the study. The participants' age ranged between 18 and 52 with an average of 28.8 (SD=9.38) and a

Table 1: Participant demographics

Group	n	Age mean	SD	% male	% female	Hours per week
Group 1	8	29.75	10.29	87.5	12.5	22
Group 2	7	28.29	11.4	71.4	28.6	30.7
Group 3	5	27.8	5.85	100	0	19.2
Total	20	28.75	9.38	85	15	24

The three groups took part in an identical test

median of 26. In total, 15% of the participants were women ($N=3$) and 85% were men ($N=17$) (Table 1). The weekly average time spent playing games was 24 hours which is just between the 22 hours reported by the Nielsen report [42] and the 25 hours from Griffiths et al.'s study [24]. They all had previous experience with games similar to the one they played during the test, and they considered open-world games and/or RPG to be one of their favorite genres.

3 RESULTS

3.1 Factor analyses

The SUS, IEQ and FSS, scores were obtained after running a factor analysis for each scale. The scores on the seven dimensions from the Attrakdiff as well as the general appreciation of the game were kept in their original form.

Oblimin rotation was preferred over Varimax as any factor in social science is most likely, to some extent, related to other factors [38]. The number of factors was selected based on parallel analysis as it is more precise than the eigenvalue rule of thumb [17, 36].

Two scores were obtained from the SUS (learnability and usability) which has been the case in other research which also found items 4 and 10 to represent learnability and not usability [7, 34, 36, 37, 50]

One single score was obtained for the IEQ after the removal of several items.

Item 10 was removed in the FSS and the two scores were obtained: absorption by activity and fluency of performance.

Cronbach's α ranged from 0.704 to 0.954. All Bartlett's tests were significant and KMO indices were above the 0.5 threshold. All factor loadings were superior to .6

3.2 Correlations

From the Attrakdiff questionnaire, inviting, likeable, good, and motivating scores were all correlated positively with appeal scores. Usability was correlated with all 7 dimensions of the Attrakdiff questionnaire with p-values ranging from .034 to .002. Appeal scores showed the highest correlation with usability, from the Attrakdiff ($r = 0.534$; $p = .002$) which confirmed the hypothesis "a". Interestingly, learnability was not correlated with any of the Attrakdiff scores. Learnability was correlated positively with absorption by activity ($r = 0.400$; $p = .020$) which was not the case for usability. The general appreciation of the game was correlated positively with immersion, the two flow scores, and all scores from the Attrakdiff except appeal.

3.3 Linear regressions

The results of the simple linear regression indicated a significant effect of usability on immersion; $B = .65$, $t(16) = 3.46$, $p < .002$, and it explained a significant proportion of variance in Immersion scores $R^2 = .43$, $F(1, 16) = 12$, $p < .002$. These results confirm the hypothesis "b". Learnability did not predict immersion scores. Usability did not predict significantly the absorption by activity scores, but it did have a significant effect on fluency of performance scores, $B = .97$, $t(18) = 4.33$, $p < .001$, and it explained 51% of variance. These results partially confirmed the hypothesis "c" as usability predicted only one of the two flow scores.

3.4 Mediations

The mediation analyses were executed based on the Baron & Kenny method [5]. The first mediation model we measured was the effect of usability on immersion mediated by appeal. As stated earlier, usability had a significant effect on immersion $B = .65$, $t(16) = 3.46$, $p = .003$ (Step 1), as well as on appeal, $B = .86$, $t(16) = 2.86$, $p = .011$ (Step 2). Step 3 of the mediation process showed that appeal had a significant effect on the level of immersion $B = .52$, $t(16) = 4.89$, $p < .001$. Step 4 revealed that a full mediation was present ($z = 2.30$, $p = .021$). The level of appeal fully mediated the effect of the perceived usability on the level of immersion experienced as the direct effect of usability on immersion was non-significant once appeal was introduced as a mediator. This confirms the hypothesis "d" (Figure 3).

A second mediation was tested to analyze the effect of usability on the general appreciation of the game, mediated by the level of immersion. Step 1 showed a significant effect of the usability level on the general appreciation $B = .94$, $t(16) = 2.51$, $p = .023$, step 2 was already verified in the first mediation model ($B = .65$, $t(16) = 3.46$, $p = .003$), and step 3 also showed a significant effect of immersion on the appreciation of the game $B = 1.51$, $t(16) = 6.17$, $p < .001$. Step 4 revealed the presence of a full mediation ($z = 2.974$, $p = .003$) as the indirect effect became non-significant when immersion was introduced as a mediator in the model. It seems to indicate that the level of perceived usability impacts immersion via the effect it has on the perceived appeal of the game. The immersion level then impacts the general appreciation of the game. All regressions in the mediation analyses met the assumption of normality and of independent errors.

4 DISCUSSION

This research aimed at better understanding the relationship between immersion, flow and usability in video games. 4 hypotheses

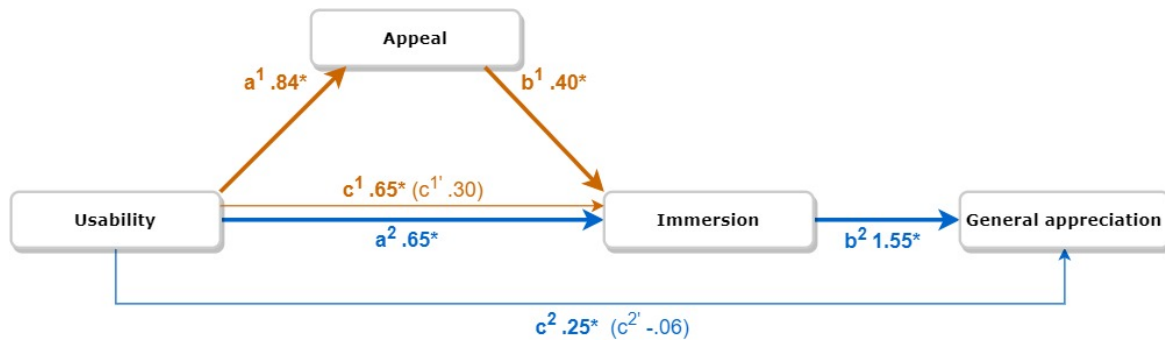


Figure 3: Mediation analyses representation. Parentheses report the p-values of the mediation analyses. * = p-value smaller than .05. Path number 1 shows the mediation effect of appeal on the relationship between usability and immersion. Path Number 2 shows the mediation effect of immersion on the relationship between usability and general appreciation.

were established before running the study. Hypothesis A was confirmed with a significant positive correlation between usability and appeal. Hypothesis B was confirmed with the positive linear regression of usability on immersion. The hypothesis C was partially confirmed as usability only predicted one of the two flow factors (fluency of performance). Hypothesis D was confirmed with a significant total mediation effect of appeal on the predictive effect of usability on immersion. Interestingly, another total mediation was also found between usability and general appreciation with the immersion score as a mediator. These results go in the same direction as previous research that has shown links between pragmatic and hedonic factors [15, 25, 27]. Other results also suggest that usability precedes appeal whilst appeal influences the immersion level [14], which is the direction of the relationship that was tested. These different research combined meant that usability had to come before appeal which in its turn would precede immersion. Furthermore, immersion lets the players have fun and enjoy the game, and hence influences the general appreciation of the game. As for flow, interestingly, only fluency of performance could be predicted by the usability scores, but not absorption by activity. The opposite results were found for learnability. These results could imply that a game that feels easy to learn will let the players be absorbed in the activity they are doing since they quickly learn how to play and navigate, whereas when the game allows the players to carry on their goals efficiently and effectively they feel more performant and capable.

The correlation between absorption by activity and fluency of performance was non-significant. The two scores based on the original calculating method (5 items for each factor without rearranging the structure based on the factor analysis results) were also not correlated. This should be investigated more in-depth to analyze if similar results are found or not with a larger sample size.

Immersion was the factor with the highest correlation with the general appreciation of the game ($r = 0.746$; $p < .001$). These two variables were also significantly correlated with the two flow scores that were used in this study.

5 CONCLUSIONS

The configuration of the tests allowed us to recreate a simulated situation that resembled a natural experience better than an artificial

experiment, in a university lab. The fact that we could make sure that participants had never played the game as well as having them play for several hours over days improved the internal validity of the study. Results showed that usability and appeal both predict immersion and general appreciation. The effect usability had on immersion was mediated by appeal, whereas the effect it had on the appreciation of the game was mediated by immersion. Usability was positively correlated with fluency of performance, whilst learnability was positively correlated with absorption by activity. Usability was correlated positively with the seven Attrakdiff scores. However, the power of the study is fairly low due to the small sample size and the results should be verified with a larger sample size. Manipulating the usability level experimentally (low, medium, high) could also provide a better understanding of the extent to which usability impacts immersion and flow, because what was measured in the present study was the subjective usability level perceived by the participants and not an objective measure of usability.

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